



LUND
UNIVERSITY

Centre for Mathematical Sciences
Division of Mathematics

Group and Ring Theory, Autumn Semester 2026

Programme 3.9 – 26.10

Starting on the 3rd of September 2026, the lectures will generally be on Mondays and Thursdays (the first week is an exception) 15:15 – 17:00 in room MH:Hörmander. Starting on the 11th of September 2026, the exercise classes will be on Fridays at 15:15 – 16:00 in MH:332B. The lecture notes and exercise sheets will be uploaded on Canvas in good time.

Optional Literature:

[B] P. B. Bhattacharya, S. K. Jain and S. R. Nagpaul, *Basic Abstract Algebra*, 2nd ed., Cambridge University Press, 1994.

Thursday 3/9	Study: Group actions (e.g. Section 5.4 to 5.4.9 in [B]) Solve: Exercise Sheet 1
Friday 4/9	Study: Applications of the Orbit-Stabiliser (e.g. Section 5.4.10 to 5.4.13 in [B]) Solve: Exercise Sheet 2
Monday 7/9	Study: Sylow theorems (e.g. Section 8.4 to 8.4.6 in [B]) Solve: Exercise Sheet 3
Thursday 10/9	Study: Applications of Sylow theorems (e.g. Section 8.4.7 to 8.5 in [B]) Solve: Exercise Sheet 4
Friday 11/9	Discuss: Exercise Sheets 1 and 2 at exercise class
Monday 14/9	Study: An introduction to modules (e.g. Section 14.1 to 14.3 in [B]) Solve: Exercise Sheet 5
Thursday 17/9	Study: Completely reducible modules (e.g. Section 14.4 in [B]) Solve: Exercise Sheet 6
Friday 18/9	Discuss: Exercise Sheets 3 and 4 at exercise class
Monday 21/9	Study: Free modules (e.g. Section 14.5 in [B]) Solve: Exercise Sheet 7
Thursday 24/9	Study: Noetherian and Artinian modules (e.g. Section 19.1 to 19.2.4 in [B]) Solve: Exercise Sheet 8
Friday 25/9	Discuss: Exercise Sheets 5 and 6 at exercise class
Monday 28/9	Study: Noetherian and Artinian rings (e.g. Section 19.2.5 to 19.2.12 in [B]) Solve: Exercise Sheet 9

Thursday 1/10	Study: Hilbert basis theorem (e.g. Section 19.2.14 to 19.2.15 in [B]) Solve: Exercise Sheet 10
Friday 2/10	Discuss: Exercise Sheets 7 and 8 at exercise class
Monday 5/10	Study: Wedderburn-Artin Theorem (e.g. Section 19.3 to 19.3.6 in [B]) Solve: Exercise Sheet 11
Thursday 8/10	Study: Smith normal form over a PID (e.g. Chapter 20 in [B]) Solve: Exercise Sheet 12
Friday 9/10	Discuss: Exercise Sheets 9 and 10 at exercise class
Monday 12/10	Study: Finitely generated modules over a PID (e.g. Section 21.1 to 21.3 in [B]) Solve: Exercise Sheet 13
Thursday 15/10	Study: Rational canonical form and tensor products (e.g. Sections 21.4 and 22.2 in [B]) Solve: Exercise Sheet 14
Friday 16/10	Discuss: Exercise Sheets 11 and 12 at exercise class
Monday 19/10	Study: Module structure of the tensor product (e.g. Section 22.3 to 22.4 in [B]) Solve: Exercise Sheet 15
Thursday 22/10	Discuss: Exercise Sheets 13, 14 and 15 at exercise class
Friday 23/10	Revision session 1 Discuss: The written exams from August 2023 and November 2024.
Monday 26/10	Revision session 2 Discuss: The written exams from August 2025 and November 2025.

The written exam will take place on the 27th of October 2026 at 8:00–13:00. The associated oral exams will take place in the afternoons of the week afterwards.

The retake written exam will take place on the 5th of December 2026 at 8:00–13:00. The associated oral exams will take place in the afternoons of the week afterwards.

More information and instructions for the oral exams will be posted on the Canvas page in due course.